

Lecture 10:

Heat flow in glaciers

Why do we care?

Glacier thermal structure

General heat equation derivation

Two examples:

1. Heat flow near the surface
2. Full thickness temperature profiles

Why do we care about heat flow in ice?

--> Because it controls ice-sheet thermal structure.

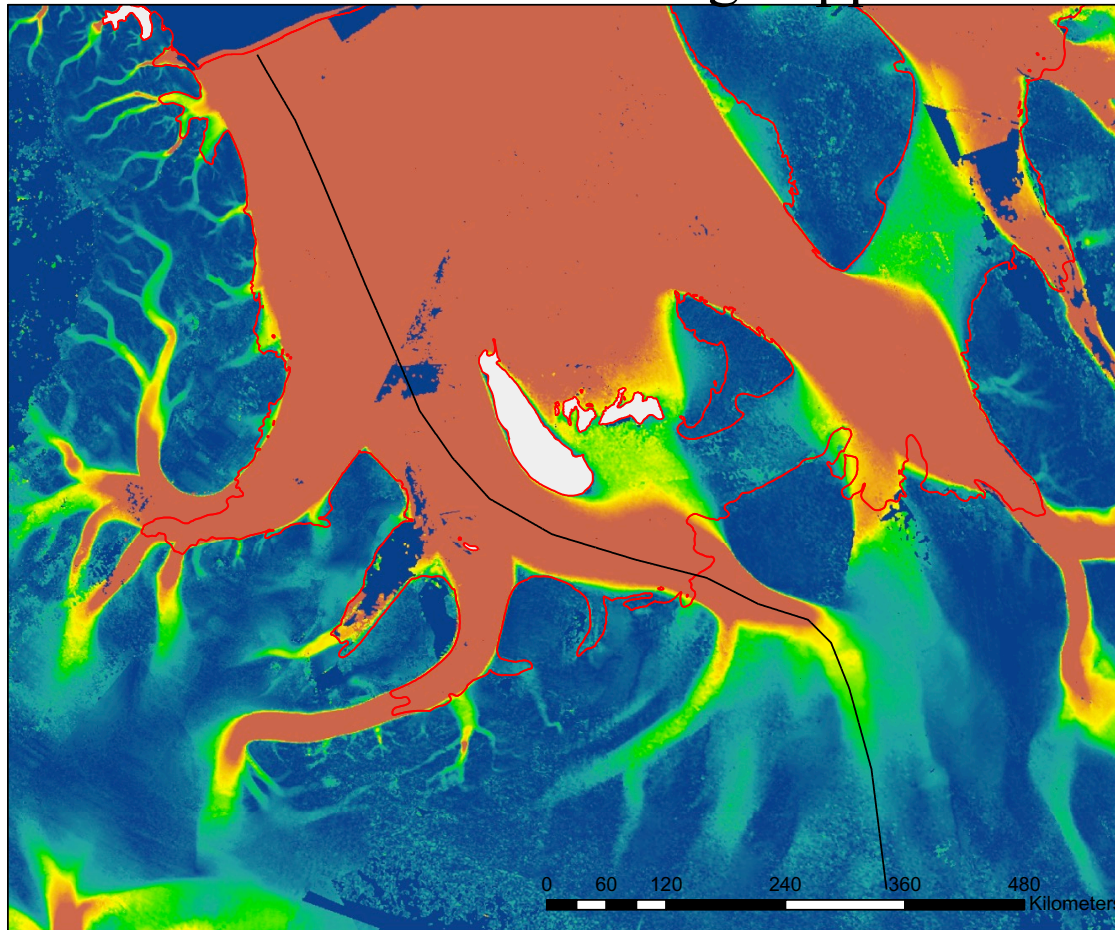
But why do we care about that?

Why do we care about heat flow in ice?

--> Because it controls ice-sheet thermal structure.

But why do we care about that?

It controls where sliding happens



Why do we care about heat flow in ice?

It controls basal melt and re-freezing rates

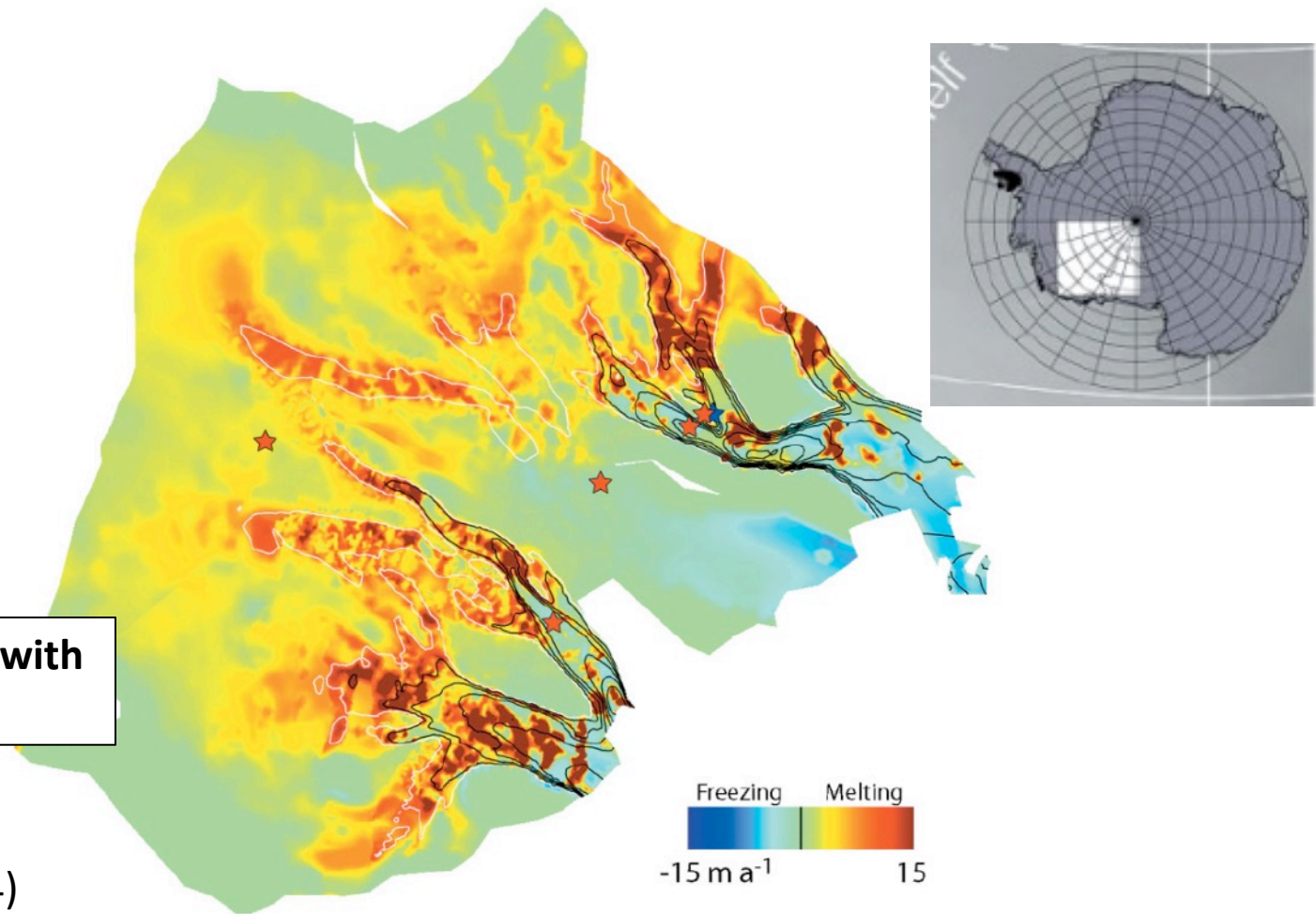
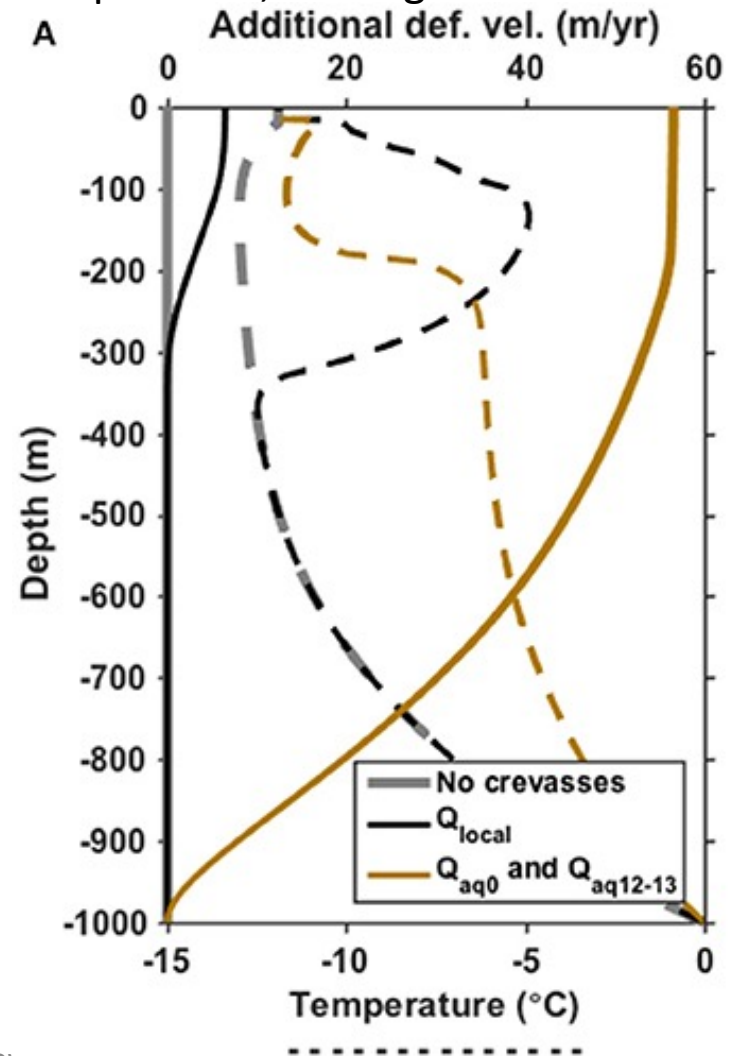
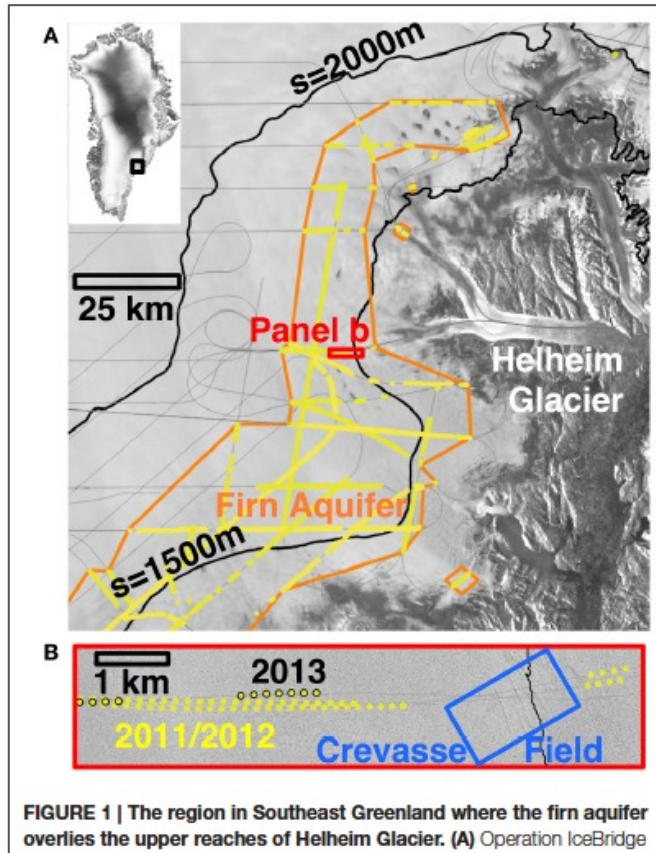


Fig. 8. Estimated basal melt/freeze rates. Flow-speed contours at 100 m a⁻¹ intervals (black) and at 50 m a⁻¹ (white) show locations of the ice streams. Stars show borehole locations discussed in the text.

Why do we care about heat flow in ice?

It controls ice viscosity

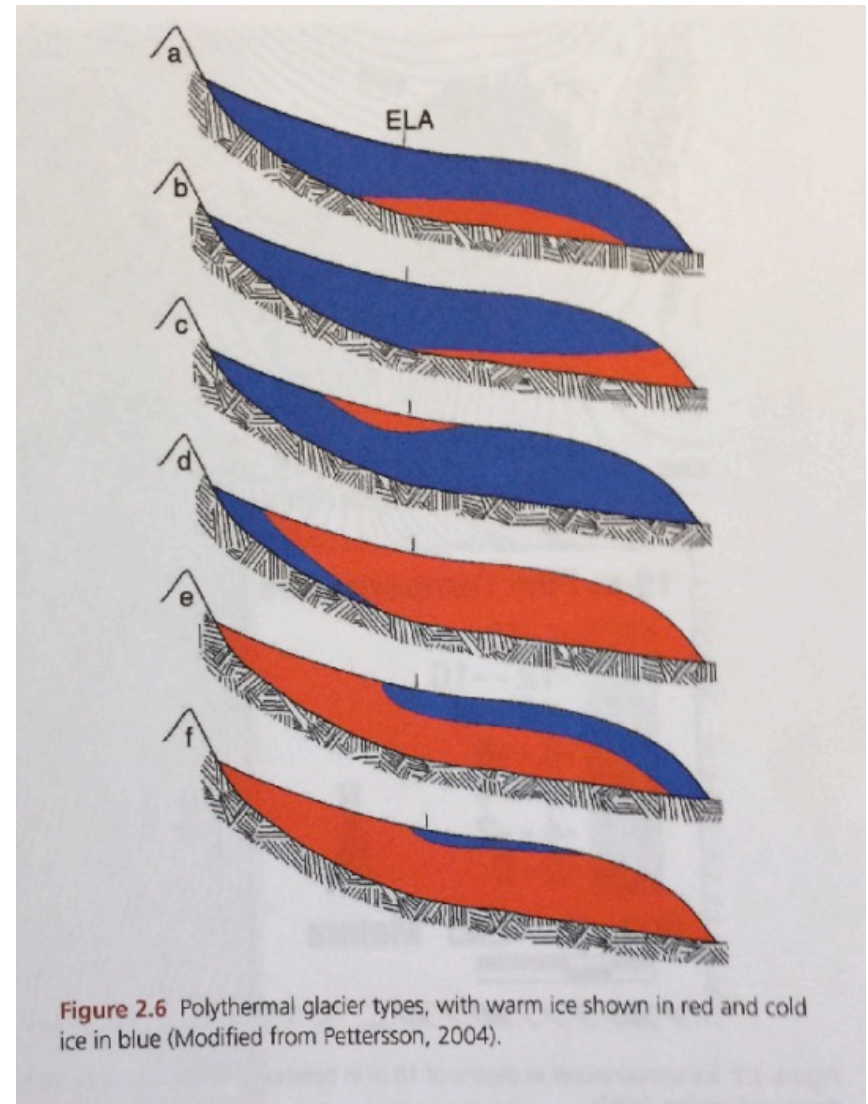
Water flows into a crevasse and warms up the ice, leading to faster flow



Poinar et al. (2017)

Glacier thermal structure

- Temperate/warm
at the melting-point
everywhere
- Polar/cold
below the melting
point everywhere
- Polythermal
A mixture



Benn and Evans (2010)

General heat flow equation.

Plan for derivation:

1. Setup the problem
2. dT/dt term
3. Diffusion term
4. Advection term
5. Bring it all together

General heat flow equation.

Plan for derivation:

1. Setup the problem
2. dT/dt term
3. Diffusion term
4. Advection term
5. Bring it all together

T is temperature
 α is thermal diffusivity
 x, y, z are dimensions
 u, v, w are components of
velocity in x, y and z directions

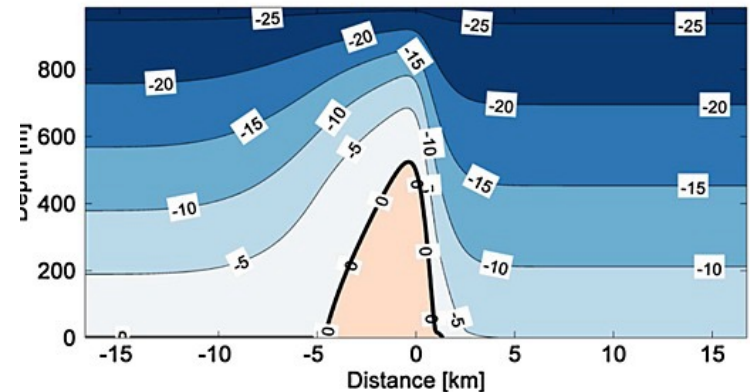
$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) - \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right)$$

What about sources of heat within the ice? (so-called source terms).

- Ice deformation
(including firn compaction).

$$\epsilon_{xx}\sigma_{xx}, \epsilon_{yy}\sigma_{yy}$$

- Refreezing of meltwater



Suckale et al. (2014)

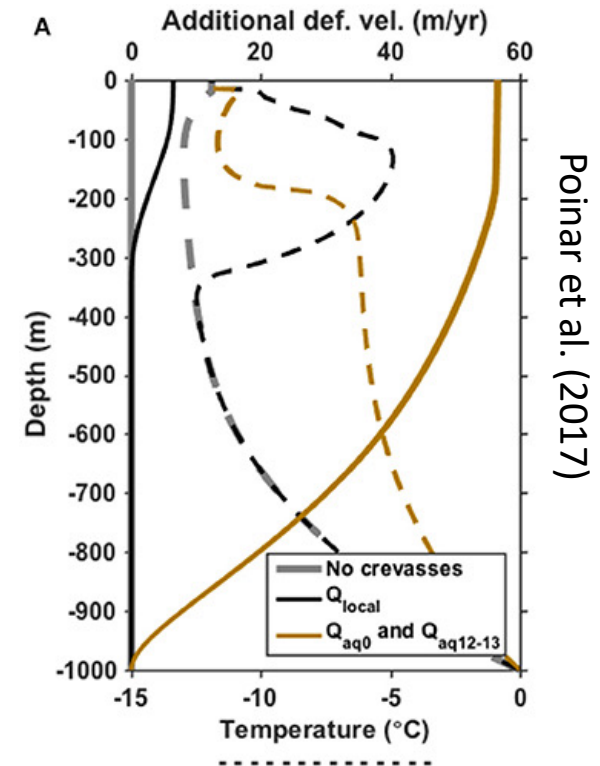
$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) - \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) + S$$

What about sources of heat within the ice? (so-called source terms).

- Ice deformation (including firn compaction).

$$\epsilon_{xx}\sigma_{xx}, \epsilon_{yy}\sigma_{yy}$$

- Refreezing of meltwater



$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) - \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) + S$$

A simple example:

it is time dependent, but ignores advection.

Example 1

Temperature variations near the surface

$$\boxed{\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) - \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) + S}$$

Example 1

Temperature variations near the surface

Let's ignore advection

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) - \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) + S$$

Example 1

Temperature variations near the surface

Let's ignore advection and source terms

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) - \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) + S$$

Example 1

Temperature variations near the surface

Let's ignore advection and source terms

$$\frac{\partial T}{\partial t} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) - \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) + S$$

and x and y directions

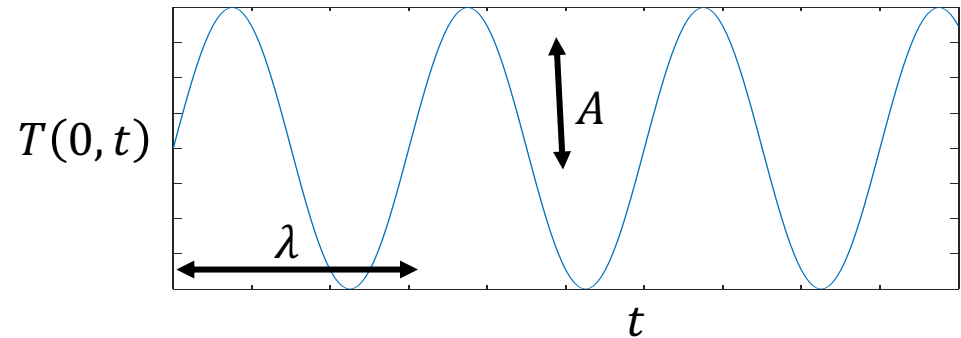
$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial z^2}$$

Example 1

Assume surface energy balance produces a sinusoidally-varying surface temperature, $T(0,t)$

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial z^2}$$

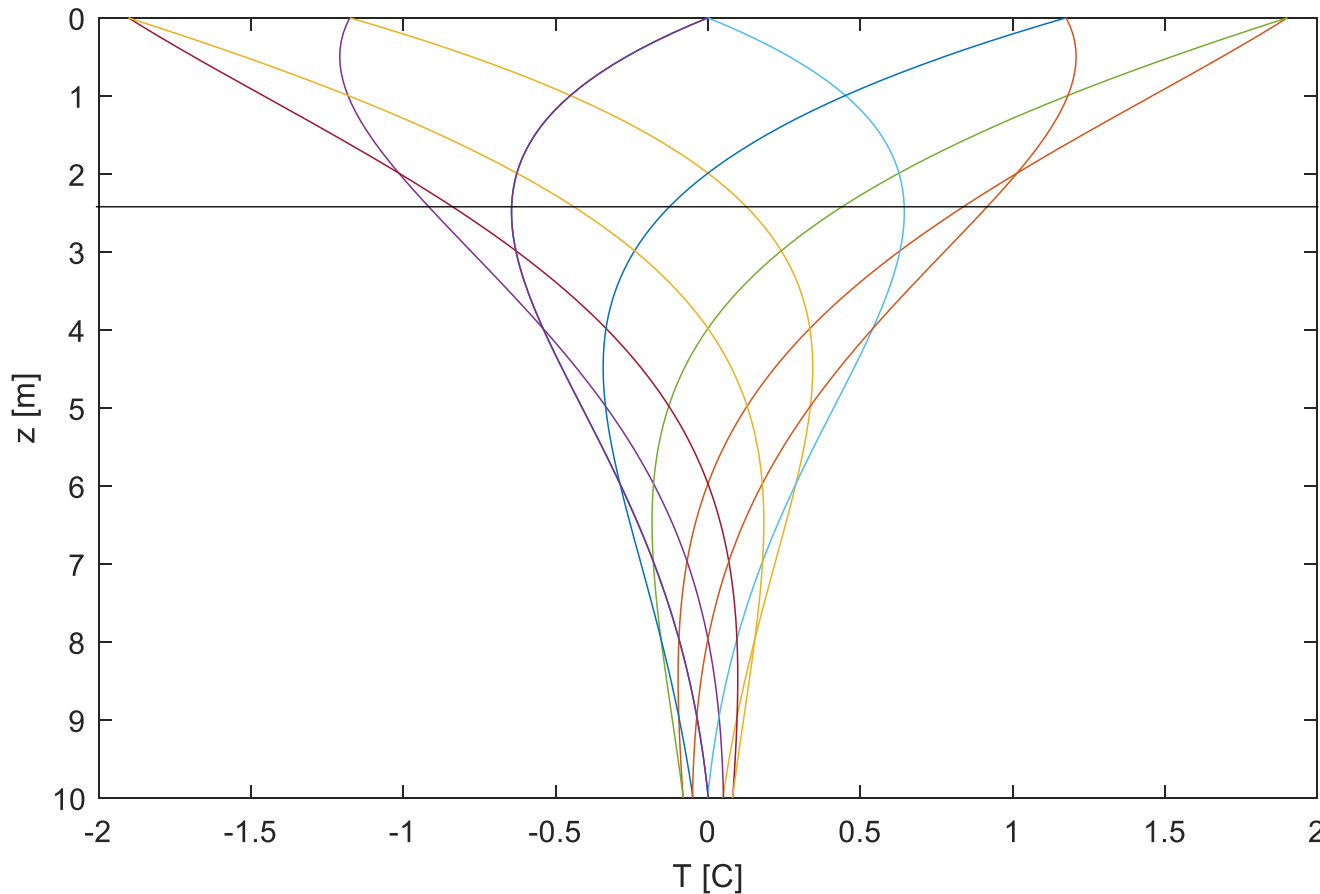
$$T(0,t) = A \sin\left(\frac{2\pi}{\lambda} t\right)$$



$$T(z,t) = A e^{-z \sqrt{\frac{\pi}{\lambda \alpha}}} \sin\left(\frac{2\pi}{\lambda} t - z \sqrt{\frac{\pi}{\lambda \alpha}}\right)$$

Example 1

$$T(z, t) = A e^{-z \sqrt{\frac{\pi}{\lambda \alpha}}} \sin \left(\frac{2\pi}{\lambda} t - z \sqrt{\frac{\pi}{\lambda \alpha}} \right)$$

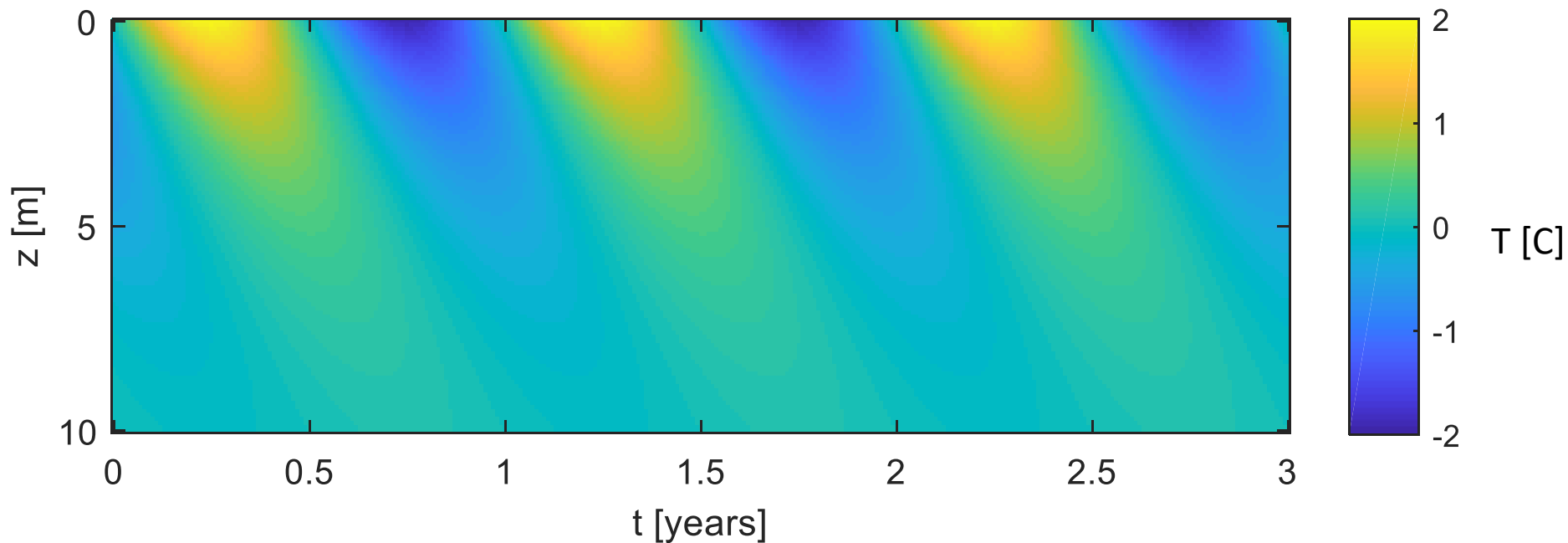


$$\sqrt{\frac{\pi}{\lambda \alpha}} = 0.3 \text{ m}^{-1}$$

Amplitude of
annual variations is
 $\sim 0.05A$ at 10 m.

Example 1

$$T(z, t) = A e^{-z \sqrt{\frac{\pi}{\lambda \alpha}}} \sin \left(\frac{2\pi}{\lambda} t - z \sqrt{\frac{\pi}{\lambda \alpha}} \right)$$



Summary

- Temperature controls where sliding happens, ice deformation, and basal mass balance.
- Heat moves around through advection and diffusion.
- Internal heat sources include strain heating, refreezing.
- Boundary conditions set by environment.
- Seasonal waves penetrate 10-20 m.

Example 2

Steady-state $T(z)$ at an ice divide

- Steady-state

$$\cancel{\frac{\partial T}{\partial t}} = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) - \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) + S$$

And again ignore source terms

$$0 = \alpha \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} \right) - \left(u \frac{\partial T}{\partial x} + v \frac{\partial T}{\partial y} + w \frac{\partial T}{\partial z} \right) + \cancel{S}$$

and x and y directions

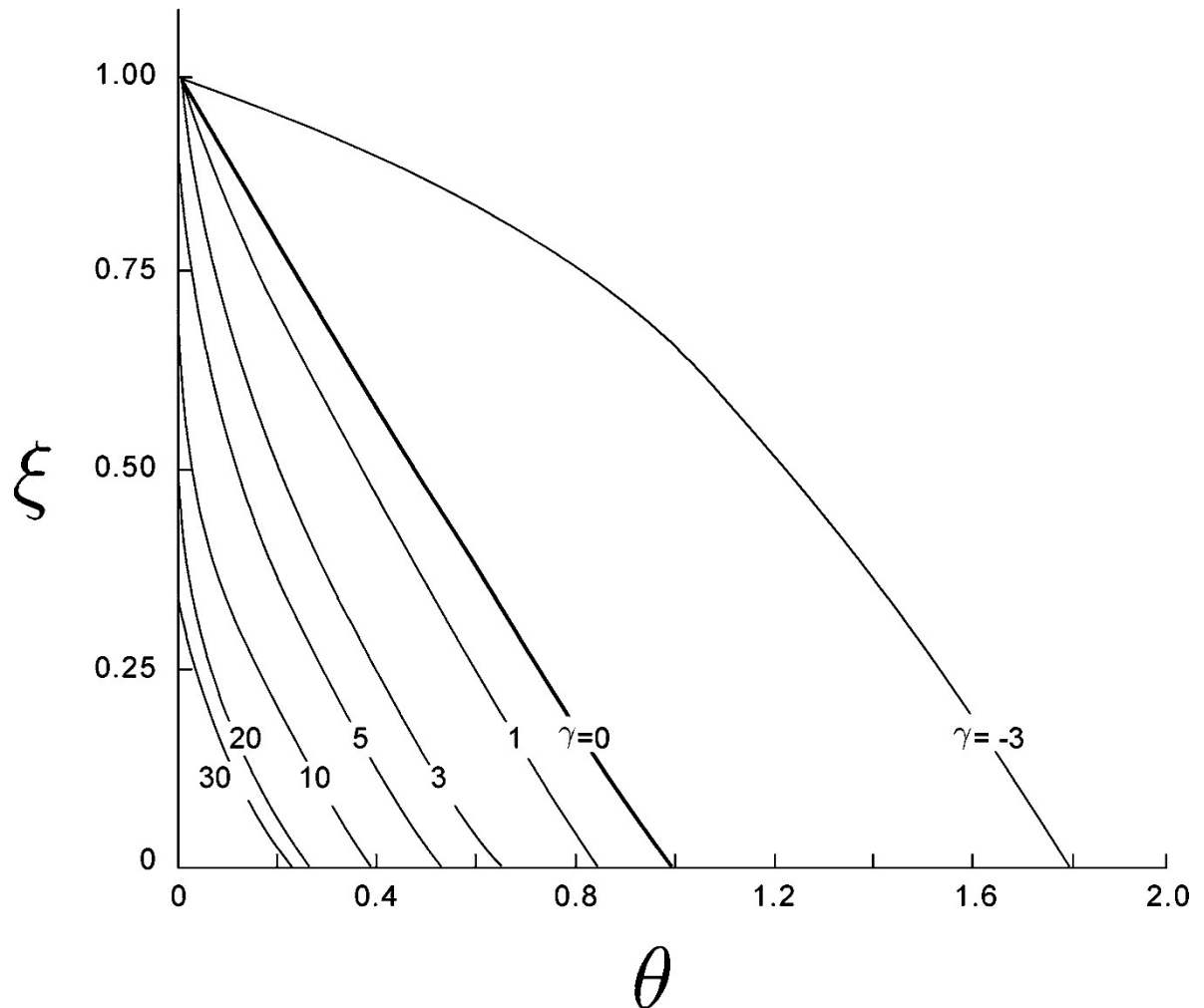
$$w \frac{\partial T}{\partial z} = \alpha \frac{\partial^2 T}{\partial z^2}$$

Example 2

Steady-state $T(z)$ at an ice divide

$$w \frac{\partial T}{\partial z} = \alpha \frac{\partial^2 T}{\partial z^2}$$

“non-dimensional”
variables $\Rightarrow$



References

- Joughin, I., Tulaczyk, S., MacAyeal, D.R. and Engelhardt, H., 2004. Melting and freezing beneath the Ross ice streams, Antarctica. *Journal of Glaciology*, 50(168), pp.96-108.
- Poinar, K., Joughin, I., Lilien, D., Brucker, L., Kehrl, L. and Nowicki, S., 2017. Drainage of Southeast Greenland firn aquifer water through crevasses to the bed. *Frontiers in Earth Science*, 5, p.5.
- Suckale, J., Platt, J.D., Perol, T. and Rice, J.R., 2014. Deformation-induced melting in the margins of the West Antarctic ice streams. *Journal of Geophysical Research: Earth Surface*, 119(5), pp.1004-1025.